NMZ Serratia genomics

Glossary for the research group. Companion to the revised November 2026 explainer.

Core terms

Term	Meaning and relevance
Isolate; strain	An organism recovered from a sample or culture; a distinguishable lineage. An isolate name is not itself a species identification.
Genotype; phenotype	Genetic constitution; an observed trait under stated conditions. Finding a candidate gene does not by itself predict the size of an effect.
Read; basecalling	A sequence fragment recorded by a sequencing instrument; conversion of nanopore electrical signal into DNA bases. A basecalled read is sequence data, not the raw electrical signal.
Assembly; contig; polishing	Reconstructing longer genomic sequences from reads; one continuous assembled sequence; refining a sequence using read evidence.
Closed circular assembly; contig break	A replicon reconstructed as a circular sequence with its ends joined; a boundary between separate assembled sequences. Closure lets a selected neighborhood be followed without a contig break, but does not independently certify every base or prove a gene's activity or product.
Chromosome; plasmid; replicon	A main genomic DNA molecule; an additional replicating DNA element; the general term for a replication unit. Plasmid identity requires supporting evidence.
bp, kb, Mb	Base pairs, thousand base pairs, million base pairs. They measure sequence length.
Annotation; CDS; predicted proteome	Inferred genome features or functions; a protein-coding sequence; the protein set predicted from annotation. The focal FAA files also contain ten short ORFs per isolate.
GC content	Percentage of DNA bases that are G or C. It is descriptive, not a degradation-performance score.
Gene; protein	The DNA sequence and its encoded product. Write the gene as italic *swrW* and the protein as SwrW in manuscript prose.
Homolog	A sequence related through shared ancestry. Homology alone does not guarantee the same substrate or function.
Pathway; substrate	A set of chemical steps; the molecule an enzyme or pathway acts on. Substrate identity matters when connecting genomic candidates to an assay.
Biosynthetic gene cluster (BGC)	A group of nearby genes that may work together to make a specialized molecule. A predicted BGC region can contain several candidate genes; a count of predicted regions is not a count of demonstrated products.
antiSMASH; region	antiSMASH is software that identifies and classifies candidate BGCs using sequence and domain rules. Its region numbers mark positions within an input sequence; matching numbers in different genomes do not establish that the regions are equivalent.
MIBiG; cluster match	MIBiG is a database of experimentally characterized BGC references. A similarity hit is evidence of resemblance; its coverage, identity and number of matching genes affect how specifically it supports a pathway hypothesis. A database label alone does not identify the product.

Term	Meaning and relevance
Biosurfactant; bioemulsifier; amphiphile	A biologically produced substance that can affect surface or interface properties; a substance that promotes or stabilizes an emulsion; a molecule with water-compatible and water-avoiding portions. These activities overlap in some molecules but are not interchangeable assay results.
Serrawettin W1; serratamolide A; serratamolide C; congener	Serrawettin W1 is also called serratamolide A. Serratamolide C is a related but chemically distinct congener. The LC-HRMS source reports putative assignments; the names should not be counted as independent discoveries when they are aliases.
LC-HRMS rows/samples 1 and 2	Chemical-analysis entries, not isolate identifiers. In the source workflows labelled glycopeptide and DCM, the reported putative assignments include serrawettin W1 (= serratamolide A) and serratamolide C. “Glycopeptide” is the workflow label, not a chemical-class assignment for these compounds. The current both-isolates coverage is a drafting assumption; the source provides no isolate key.
EPS; CPS; capsule; biofilm	EPS here means extracellular polymeric substances: a matrix that may include polysaccharides, proteins and DNA. CPS means capsular polysaccharide associated with the cell. A biofilm is organized microbial growth often associated with a matrix. These terms do not identify one chemical product.
PNAG	Poly-β-1,6-N-acetylglucosamine, a matrix/adhesion polysaccharide associated with the *pga* system. A lone *pgaB* annotation does not establish a complete local *pga* system.
Cellulose; bcs	A β-1,4-linked glucose polymer; genes associated with bacterial cellulose synthesis, processing and export. Focal *bcs* loci are genomic candidates, not cellulose production measurements.
Gene cluster; operon	Nearby genes implicated in a shared function; genes transcribed together as a unit. A predicted region or conserved neighborhood is not automatically an experimentally established operon.
Hydrocarbon; aromatic compound	A molecule containing only carbon and hydrogen; a compound with aromatic ring chemistry. Oxygenated aromatics can be aromatic without being hydrocarbons.
Alkane; oxidation; monooxygenase	A saturated hydrocarbon; a chemical oxidation step; an enzyme that incorporates one oxygen atom into a substrate in the relevant reaction. An enzyme-family label does not identify the tested substrate.
Phylogeny; branch support	Inferred sequence relationships; support for a grouping under the analysis. A node with 99% support is not 99% certainty of biochemical activity.
Gene neighborhood; synteny	Local gene arrangement; conserved gene organization in the comparative usage here. Proximity does not prove co-regulation or a shared substrate.
LC-HRMS; MS/MS	Liquid chromatography-high-resolution mass spectrometry; fragmentation of selected ions to help distinguish structures. Exact mass alone can fit multiple compounds; report assignments at the confidence supported by the analytical evidence.
Tentative annotation; confirmed identity	A candidate chemical label; an identification supported by appropriate reference and standard evidence. Genomics can support plausibility without upgrading analytical confidence.
Peak-area percentage	Relative detector response under a stated analysis and denominator. It is not automatically concentration, purity, mass fraction or yield.

Term	Meaning and relevance
Surface tension; interfacial tension	Physical properties at a liquid-gas or two-phase interface. Report the actual interface and measurement method.
Emulsification index; E24	A protocol-dependent estimate of emulsion formation or stability, often the emulsion-layer height divided by total liquid-column height after 24 hours. It is not the same as tension reduction.
Pour point; PPD	A low-temperature flow endpoint under a defined test; pour-point depressant. A lower pour point does not by itself establish chemical degradation.
Transformation; mineralization	Chemical conversion of a substrate; conversion to inorganic end products. Apparent disappearance alone establishes neither.
Virulence-associated homolog	A match to a protein implicated in a pathogenic process in a reference context. It does not measure the focal isolate's pathogenicity or establish safety.

Technical terms - optional reading

Term	Meaning and relevance
ONT; WGS; FASTQ; FASTA	Oxford Nanopore Technologies; whole-genome sequencing; sequence plus per-base quality values; sequence without those quality fields.
Q score; N50; sequencing depth	A logarithmic quality estimate; the length at or above which half the assembled bases occur in sequences that long or longer; mapped read representation per position. None alone proves assembly accuracy.
ANI; AF; dDDH	Average nucleotide identity across aligned genome regions; fraction of sequence aligned under the method; computational estimate of DNA-DNA hybridization relatedness. ANI is not identity across every base of the whole genome. Species guides depend on the method and reference set.
GTDB; GTDB-Tk; TYGS	Genome Taxonomy Database; its classification toolkit; Type (Strain) Genome Server, a web service using type-strain comparisons, GBDP distances, dDDH and a phylogram. They are systems that use relatedness evidence, not names of equivalent metrics.
ANI screening; species radius	Classification against reference genomes using ANI and aligned fraction; the representative-specific identity boundary used by GTDB. The focal queries were classified by screening, without marker-tree placement.
GBDP	Genome BLAST Distance Phylogeny: genome-comparison methods used by TYGS. The d4 formula for focal dDDH estimates is distinct from the d5 distances used for the tree.
Type strain; representative; LPSN	Strain anchoring a species name; database comparison genome; nomenclature resource. A representative need not be the type strain.
Ortholog; paralog; orthogroup	Homologs related through speciation; through duplication; an inferred group under a method's rules. Conserved function is not guaranteed.
ProteinOrtho; PoFF	Protein grouping method; its gene-order extension. Keep sequence-based and gene-order outputs distinct.
Identity; alignment coverage; E value	Exact residue matches over an alignment; fraction of a specified sequence covered; expected chance matches in the search space. State the denominator and threshold.
HMM; Pfam	Statistical profile used to identify a protein family; a protein-family resource. A profile match supports a family assignment, not necessarily a particular substrate or activity.

Term	Meaning and relevance
NRPS; C-A-PCP-TE	Nonribosomal peptide synthetase; condensation, adenylation, carrier and thioesterase domains. Modules can be used iteratively; domain labels alone do not determine the complete product.
Condensation_Starter	Specialized N-terminal condensation domain (C_Starter) in NRPS enzymes (e.g. SwrW residues 1-298 with the conserved HHLIADG motif) that attaches a fatty-acyl chain directly to the first amino acid without requiring an upstream carrier protein.
Stachelhaus code	The 10 conserved amino acid residues lining the substrate-binding pocket of an NRPS adenylation (A) domain that predict which amino acid is activated. In SwrW, the DVWHVSLVDK motif confers 100% specificity for L-serine.
ProtoToRegion_RiQ	antiSMASH metric comparing an identified protocluster to a reference BGC in MIBiG. Combines sequence identity, gene synteny/order, and component overlap into a 0.00-1.00 score; a single matching gene can yield an inflated score if order is non-applicable.
yersinopine (CntM/CntL)	An opine-like metallophore / nicotianamine-related metal chelator synthesized by CntM (nicotianamine synthase) and CntL (dehydrogenase) enzymes, detected uniquely in C14-NMZ Region 7 with a 0.84 MIBiG match (BGC0002490.2).
ranthipeptide	Radical non-alpha-thioether peptide, a class of RiPP natural products characterized by radical SAM enzyme-mediated crosslinks. Present uniquely in C14-NMZ Region 3 with TIGR04085 / PF04055 radical SAM and PqqD precursor/recognition architecture.
pig; swrW	The *pig* locus is associated with prodigiosin biosynthesis; *swrW* encodes the large SwrW nonribosomal peptide synthetase associated with serrawettin W1/serratamolides. The focal locus and reported chemistry support a candidate relationship, not a gene-specific causal test.
Gap1-like NRPS; stephensiolide	Gap1-like describes a comparative NRPS similarity/architecture. The focal SM and C14 Region 10 NRPS architectures differ from one another; that comparison alone does not assign a putative LC-HRMS stephensiolide label to either locus.
hpa; paa; pob/pca	Labels for hydroxyphenylacetate-, phenylacetate- and hydroxybenzoate/protocatechuate-associated machinery in this interpretation. Use locus and pathway evidence rather than gene names alone.
4-HBA; PobA; PcaK	4-hydroxybenzoate; a hydroxylase acting on it in characterized systems; an aromatic-acid transporter. A downstream PobA-like hit does not establish upstream steps of a complete toluene route.
wzb/wzc; HADEG “Emulsan” category	Generic capsule-associated phosphatase/kinase markers. Their database category does not determine the chemical structure or activity of a polymer produced by NMZ.
LadA; FMN; LLM	A monooxygenase-associated family name; flavin mononucleotide cofactor; luciferase-like monooxygenase terminology. Here LLM refers to an enzyme family; a LadA-like hit does not by itself establish alkane hydroxylation.
A0ABS9YXJ5	Identifier for the comparison sequence labeled “Candidatus Mycolicibacterium alkanivorans NtaA/DmoA FMN monooxygenase” in the LadA-like tree. It is an outgroup comparison, not an experimental result from NMZ.
AlkB naming ambiguity	DNA-repair AlkB and membrane alkane-hydroxylase AlkB are different proteins. The focal annotations concern the former.

Term	Meaning and relevance
Adduct; mass error; authentic standard	An ion associated with another species such as H^+ or Na^+; measured versus expected mass difference; a reference compound measured for comparison.
ShlA-ShlB; Has; T6SS; VFDB	Hemolysin/secretion-activation pair; heme-acquisition system; type VI secretion system, often involved in competition; Virulence Factor Database. Multiple labels may match the same locus.
Causality; controls; replicate	Evidence that changing a factor changes an outcome; comparisons addressing alternative explanations; independently repeated units whose biological or technical level must be defined.
ProteinOrtho; orthologous group; core, shared and genome-specific groups	ProteinOrtho clusters similar proteins from several genomes into groups. A core group has a member in every genome, a shared group in some genomes, and a genome-specific group in one. Groups are sequence-based, so they show relatedness, not function, and a single-genome group can reflect a divergent gene or an annotation difference.
m/z; adduct; [M+H]+; ppm error	Mass-to-charge ratio measured by the mass spectrometer; an ion formed when the molecule picks up a proton or a metal ion (H+, Na+, K+). [M+H]+ is the molecule plus one proton. Parts-per-million error compares the observed and calculated m/z. A match supports a formula, not a unique structure.
Depsipeptide; lactone; congener series	A peptide-like ring that contains ester bonds as well as amide bonds; an ester inside a ring. Serratamolide A (= serrawettin W1) is a 14-membered ring of two 3-hydroxydecanoyl-serine units. Congeners are related compounds, often differing in acyl-chain length (each extra CH2 adds 14.0157 Da).